

NSE Expiry Day Dynamics & VWAP Settlement Anomalies

Research Methodology Workflow, Processing Steps & Complete Hypotheses (H1–H30)

1 Methodology & Rough Processing Steps (Pipeline Overview)

This research examines the microstructure of 10 Nifty 50 stocks (5 liquid, 5 illiquid) and their corresponding futures contracts on the National Stock Exchange (NSE) during the final 30-minute settlement window (15:00–15:30 IST) across 12 Monthly Expiry Thursdays and 12 matched Control trading days in 2022. Below is the step-by-step processing workflow:

Step 1: Sample Selection & Experimental Design (10 Symbols x 24 Dates)

- Target Asset Universe: Select 5 Liquid Nifty 50 stocks (RELIANCE, TCS, INFY, HDFCBANK, ICICIBANK) and 5 Illiquid Nifty 50 stocks (GRASIM, HEROMOTOCO, SHREEVECEM, UPL, TECHM).
- Experimental Pairing: Match 12 Monthly Expiry Thursdays in 2022 against 12 Control Thursdays (typically 2 weeks prior to expiry) to isolate expiry-day effects from day-of-week seasonality.
- Settlement Focus Window: Extract tick-by-tick cash market and derivatives market data during the NSE VWAP settlement window (15:00 to 15:30 IST).

Step 2: High-Speed Binary Tick Data Parsing (Stage 1)

- Input Data Format: Ingest fixed-width binary text files (.DAT.gz) from NSE for CASH.Orders, CASH.Trades, FAO.Orders, and FAO.Trades.
- Predicate Pushdown: Filter raw strings on symbol substrings and instrument types ('EQ' for Cash, 'FUTSTK' for Futures) before extracting schema columns.
- Column Extraction: Extract 17 standardized fields including Order/Trade numbers, timestamps (Jiffies), prices, volumes, client identity, and algo indicator flags, saving as partitioned Parquet.

Step 3: Temporal & Microstructure Enrichment (Stage 2)

- Timestamp Normalization: Convert NSE Jiffies (1/65536th of a second since epoch) into standard Python/Spark timestamps and HH:mm:ss time buckets.
- Currency & Flag Mapping: Convert Paise to INR Rupees (/ 100.0), flag the 30-minute settlement window (15:00-15:30 IST), and identify Expiry vs. Control dates.
- Participant & Algo Tagging: Categorize Client Identity into Custodian (Institutional), Proprietary, and NCNP (Retail), and classify Algo Indicator flags into Algo vs. Non-Algo execution.
- Dynamic Partitioning: Save enriched datasets partitioned by ['symbol', 'trade_date'] to enable instant single-symbol filter pushdown during downstream CLOB replay.

Step 4: Trade-Level Microstructure Metrics Computation (Stage 3)

- A1 (VWAP Trajectory): Compute cumulative and rolling 1-minute Volume-Weighted Average Price (VWAP) trajectories across 15:00-15:30 IST.
- A2 (Basis Divergence): Compute intraday Futures-Cash basis spread and Basis Volatility (standard deviation of basis).

- A3-A7 (Participant & Algo Profiling): Quantify Prop/Custodian trade shares, Algo volume contribution, Immediate-Or-Cancel (IOC) aggressiveness, cancellation-to-trade ratios, and iceberg hidden order volume.

Step 5: High-Performance Limit Order Book (CLOB) Reconstruction (Stage 4)

- Chronological Event Merging: Merge order entry/modify/cancel events with trade execution events into a unified chronological stream.
- Data-Oriented Design (C++): Replay events through a cache-line aligned C++ order book engine (FlatPriceBook with $O(\log N)$ price level lookup and zero heap fragmentation).
- 1-Second Depth Snapshots: Emit Top-10 bid/ask price levels and available shares at every second from 15:00:00 to 15:30:00 IST (1,800 snapshots per symbol/date).

Step 6: Limit Order Book Liquidity & Price Impact Analysis (Stage 5)

- B1-B2 (Spread & Depth Dynamics): Calculate percentage bid-ask spread widening and Top-5 / Top-10 order book depth erosion across the settlement window.
- B3-B5 (OFI, Kyle's Lambda & Asymmetry): Calculate Order Flow Imbalance (OFI), estimate Kyle's Lambda (price impact per million INR traded via OLS regression), and measure Bid-vs-Ask book asymmetry.

Step 7: Bloomberg Terminal Integration & Cross-Market Validation (Stage 6)

- C1 (Roll Pressure): Ingest Bloomberg calendar spread data to quantify institutional rollover urgency and predict net directional punch ('UP' or 'DOWN').
- C2-C3 (Cost-of-Carry & Cross-Validation): Calculate implied repo rates and test whether Bloomberg roll direction predicts actual NSE Cash VWAP terminal drift and CLOB book imbalance.
- C4 (Event Study): Evaluate price trajectories around block trade executions during the final 15 minutes.

Step 8: Formal Hypothesis Testing Engine & Statistical Reporting (Stage 7)

- Statistical Test Execution: Evaluate all 30 formal hypotheses (H1-H30) using Paired t-tests, Wilcoxon Signed-Rank tests, Mann-Whitney U tests, OLS regression, Spearman rank correlation, and Binomial tests.
- FDR Multiple-Testing Correction: Apply Benjamini-Hochberg False Discovery Rate correction across all p-values to control Type I errors.
- Artifact Compilation: Automatically generate publication-quality figures (fig1 to fig10), summary statistical tables, and the final Markdown research paper.

Step 9: Modular Staged Execution & Scale-Up Strategy

- Staged Workflow: Start processing with 1 or 2 symbols/dates for rapid initial verification of the entire pipeline.
- Incremental Scaling: Copy additional raw data files into 'data/raw/' at any time; pipeline automatically skips previously parsed/built snapshots and dynamically updates partitions without wiping existing data.

2 Complete List of Tested Hypotheses (H1 – H30)

The table below outlines all 30 formal statistical hypotheses evaluated in Stage 7 of the pipeline, along with their null hypothesis (H_0), alternative hypothesis (H_1), and statistical testing methodology.

ID	Hypothesis Name	Null Hypothesis (H_0)	Alternative Hypothesis (H_1)	Statistical Test
H1	Expiry Day Basis Volatility Elevation	Basis volatility is identical on Expiry and Control days	Basis volatility is significantly higher on Expiry days due to VWAP settlement trading	Paired t-test & Wilcoxon Signed-Rank
H2	Liquidity Impact on Expiry Basis Volatility	Illiquid and Liquid stocks experience identical basis volatility elevation	Illiquid stocks exhibit significantly higher basis volatility on Expiry day than Liquid stocks	Mann-Whitney U Test
H3	Expiry vs Control Spread Widening	Bid-Ask spreads remain unchanged between Expiry and Control days	Bid-Ask spreads widen significantly during the 15:00-15:30 window on Expiry day	Paired t-test & Wilcoxon Signed-Rank
H4	Spread Widening in Illiquid vs Liquid Symbols	Spread widening on Expiry day is equal across liquidity groups	Illiquid symbols experience significantly greater percentage spread widening than Liquid symbols	Mann-Whitney U Test
H5	CLOB Top-5 Depth Erosion on Expiry Day	Available order book depth at Top-5 price levels is unchanged on Expiry day	Top-5 limit order book depth erodes significantly on Expiry day	Paired t-test & Wilcoxon Signed-Rank
H6	Asymmetric Depth Erosion (Illiquid vs Liquid)	Depth erosion on Expiry day is identical across liquid and illiquid stocks	Illiquid symbols suffer more severe percentage depth erosion than Liquid symbols	Mann-Whitney U Test
H7	Final 10-Minute Depth Collapse	LOB depth remains stable across the entire 30-minute settlement window	LOB depth drops significantly in the final 10 minutes (15:20-15:30 IST) of Expiry day	Paired t-test & Wilcoxon Signed-Rank
H8	Cash Order Flow Imbalance (OFI) Expiry Elevation	Absolute Cash OFI is identical on Expiry and Control days	Absolute Cash OFI is significantly elevated on Expiry day due to directional settlement pressure	Paired t-test & Wilcoxon Signed-Rank
H9	Futures OFI Expiry Elevation	Absolute Futures OFI is identical on Expiry and Control days	Absolute Futures OFI is significantly elevated on Expiry day due to contract rolling/closing	Paired t-test & Wilcoxon Signed-Rank
H10	Cross-Market OFI Synchronization	Cash and Futures OFI are uncorrelated during the settlement window	Cash and Futures OFI exhibit strong positive correlation on Expiry day	Pearson & Spearman Correlation

ID	Hypothesis Name	Null Hypothesis (H_0)	Alternative Hypothesis (H_1)	Statistical Test
H11	Kyle's Lambda (Price Impact Elevation)	Price impact per million INR traded is identical on Expiry and Control days	Kyle's Lambda (price impact) is significantly higher on Expiry day	Paired t-test & Wilcoxon Signed-Rank
H12	Illiquid Price Sensitivity	Kyle's Lambda elevation is uniform across all Nifty 50 symbols	Illiquid symbols show a significantly larger increase in Kyle's Lambda on Expiry day	Mann-Whitney U Test
H13	Proprietary Trader Activity Elevation	Proprietary trader share of trading volume is unchanged on Expiry day	Proprietary trader share of volume is significantly higher on Expiry day	Paired t-test & Wilcoxon Signed-Rank
H14	Custodian Trade Share Stability	Custodian institutional trading volume share changes dramatically on Expiry day	Custodian trade share shows minimal deviation between Expiry and Control days	Paired t-test & Wilcoxon Signed-Rank
H15	Algorithmic Order Contribution Elevation	Algorithmic order entry share is identical on Expiry and Control days	Algorithmic order entry share is significantly higher during Expiry settlement	Paired t-test & Wilcoxon Signed-Rank
H16	Algo IOC Aggressiveness Elevation	Immediate-Or-Cancel (IOC) order usage by algos is identical across days	Algorithmic IOC usage increases significantly on Expiry day for urgent execution	Paired t-test & Wilcoxon Signed-Rank
H17	Expiry Day Order Cancellation Ratio Elevation	Order cancellation-to-execution ratio is identical on Expiry and Control days	Order cancellation ratio is significantly higher on Expiry day due to fleeting liquidity	Paired t-test & Wilcoxon Signed-Rank
H18	High-Frequency Quote Modification Intensity	Order modification frequency per minute is unchanged on Expiry day	Quote modification frequency is significantly elevated on Expiry day	Paired t-test & Wilcoxon Signed-Rank
H19	Iceberg Order Hidden Volume Contribution	Hidden iceberg order volume is identical on Expiry and Control days	Hidden iceberg order volume is significantly higher on Expiry day	Paired t-test & Wilcoxon Signed-Rank
H20	Bloomberg Roll Direction Predicts Cash VWAP Drift	Bloomberg roll pressure direction has no predictive power over terminal VWAP drift (<50% accuracy)	Bloomberg roll direction correctly predicts the sign of cash VWAP drift (>50% accuracy)	Binomial Test (H0: p=0.5)
H21	Roll Intensity Correlation with Magnitude of Cash VWAP Drift	No correlation exists between roll intensity and magnitude of cash VWAP drift	Roll intensity is positively correlated with absolute cash VWAP terminal drift	Spearman Rank Correlation
H22	Roll Direction Predicts Final CLOB Book Asymmetry	Bloomberg roll direction does not predict final limit order book imbalance (<50% accuracy)	Bloomberg roll direction predicts whether final CLOB imbalance is bid-heavy or ask-heavy	Binomial Test (H0: p=0.5)

ID	Hypothesis Name	Null Hypothesis (H_0)	Alternative Hypothesis (H_1)	Statistical Test
H23	Cost of Carry Mispricing Correlation with Cash OFI	No correlation exists between basis mispricing and Cash Order Flow Imbalance	Absolute cost-of-carry mispricing correlates positively with Cash OFI	Spearman Rank Correlation
H24	Cash VWAP Terminal Acceleration	VWAP drift is linear across the 30-minute settlement window	VWAP drift accelerates significantly in the final 10 minutes (15:20-15:30 IST)	Paired t-test & Wilcoxon Signed-Rank
H25	Futures Volume Concentration in Final 15 Minutes	Futures trading volume is evenly distributed across the 15:00-15:30 window	Futures volume concentrates significantly in the 15:15-15:30 IST window	Paired t-test & Wilcoxon Signed-Rank
H26	Expiry Day Turnover Elevation	Combined Cash and Futures INR turnover is identical on Expiry and Control days	Combined INR turnover is significantly elevated on Expiry day	Paired t-test & Wilcoxon Signed-Rank
H27	Cross-Market Spread Convergence at 15:30 IST	Futures-Cash basis spread at 15:30 IST is identical on Expiry and Control days	Futures-Cash basis converges closer to zero at 15:30 IST on Expiry day	Paired t-test & Wilcoxon Signed-Rank
H28	Algorithmic Share of Trade Volume Elevation	Algo share of executed trade volume is unchanged on Expiry day	Algorithmic execution volume share increases significantly on Expiry day	Paired t-test & Wilcoxon Signed-Rank
H29	Order Book Imbalance Persistence	Intraday autocorrelation of OFI is identical on Expiry and Control days	OFI autocorrelation is significantly stronger on Expiry day due to directional push	Paired t-test & Wilcoxon Signed-Rank
H30	Expiry Day Intraday Price Reversal Post-15:15 IST	Price movement post-15:15 is independent of 15:00-15:15 movement	Prices exhibit significant mean-reverting reversal post-15:15 IST on Expiry day	Paired t-test & Wilcoxon Signed-Rank